PRODUCT DEMO · INTERNAL WALKTHROUGH

AI Workforce

Conversational AI voice agents for health-plan & PBM member services

An always-on AI Customer Service Representative (CSR) that answers member calls by phone, resolves formulary, benefits, claims, prior-authorization, and pharmacy-network inquiries autonomously, and hands off to a live agent the moment it can't. This document walks through what every screen does, the CSR scenario it solves, the mechanics behind it, and the cost/resolution math behind the numbers on the dashboard.

BETA v0.1 — active pilot build, not yet production-certified

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Internal product walkthrough

01 — EXECUTIVE SUMMARY

Why this exists

THE PROBLEM IN MEMBER SERVICES

- High call volume on repetitive, structured inquiries: coverage checks, prior authorization (PA) status, in-network pharmacy lookups, claim status, benefit explanations.
- A fully-loaded human-handled contact-center call runs roughly $10–15, and cost scales linearly with headcount.
- Quality varies analyst-to-analyst; QA review is manual and only samples a fraction of calls.
- Coverage gaps outside business hours, and long queue times during peak call windows.

WHAT AI WORKFORCE DOES ABOUT IT

- LLM voice agent with function/tool-calling against live PBM data (eligibility, claims, formulary, pharmacy network) — not a scripted IVR tree.
- Retrieval-augmented generation (RAG) over your own uploaded plan documents, with citations, so answers are grounded rather than hallucinated.
- A supervisor cockpit (Live Operations) with real-time confidence scoring and one-click human takeover — nothing runs fully unsupervised.
- Every call is scored like a QA analyst would score it (accuracy, compliance, empathy, resolution), automatically, on 100% of calls instead of a sample.

WHAT'S IN THIS DOCUMENT

Section 2 explains the technical architecture end-to-end. Section 3 walks every screen in the product — Dashboard, Agent Studio, Live Operations, Conversations, Customers, Workflows, Analytics, AI Training Center, and Data Import — each with its CSR use case, how it works under the hood, the business value, and a table of every button and what it does. Section 4 lists the 12 actions ("tools") the AI can take on a call. Section 5 explains the cost-saved and resolution-rate math behind the dashboard tiles, formula by formula. Section 6 covers data handling. Section 7 is the honest beta disclaimer and roadmap.

02 — ARCHITECTURE

How it works, under the hood

The technical pipeline behind every call, from the phone ringing to the transcript being scored.

1. **Call ingress.** Twilio routes the inbound PSTN call to a webhook. The backend opens a `Call` record (Twilio call SID, from/to numbers, status = in_progress) and starts a conversation session.
2. **Turn loop.** On each turn, caller speech is converted to text (ASR), appended to the transcript, and sent to the LLM together with the active agent's persona/department prompt and a function-calling schema listing the 12 registered tools. The model either answers directly or calls a tool; the reply is spoken back via that agent's configured text-to-speech (TTS) voice.
3. **Grounding via RAG.** For coverage/policy questions the model calls `search_documents`, which runs a vector-embedding similarity search over PDF/DOCX chunks uploaded per-agent in Agent Studio and returns cited passages (filename, page, confidence) instead of an unsourced answer.
4. **Live data access.** For member-specific facts — eligibility, claim status, formulary tier/copay, in-network pharmacies — the model calls typed tool handlers that query the PBM data layer directly, the same source of truth a human CSR would look up.
5. **Per-turn confidence.** Every assistant message is scored 0–100% and streamed to Live Operations in real time, so a supervisor can see the AI is unsure before the caller notices.
6. **Guardrails.** The model can call `escalate_to_human`, `schedule_callback`, or `create_ticket` itself. A supervisor can also force a takeover at any moment from Live Operations — an `ai_paused` flag is checked at the top of every turn, so the AI goes silent without dropping the call.
7. **Post-call processing.** Once the call ends, background jobs generate a structured summary, classify sentiment, set `resolution_status` (resolved / escalated / abandoned), and run an LLM-as-judge QA pass scoring accuracy, compliance, empathy, and resolution — each 0–100, with written notes.
8. **Storage & multi-tenancy.** Postgres persists calls, per-turn messages, tool-call traces, customer profiles, and appointments, all scoped by tenant. The Angular frontend authenticates via JWT and reads/writes through the FastAPI REST layer.

RAG (retrieval-augmented generation) — grounding LLM answers in retrieved document chunks instead of the model's unaided memory.

ASR / TTS — automatic speech recognition (voice→text) and text-to-speech (text→voice).

Function / tool calling — the LLM invoking a typed backend action mid-conversation instead of only generating text.

LLM-as-judge — using a separate model pass to score a transcript against a rubric, the way a QA analyst would.

Embeddings — numeric vector representations of text used for semantic similarity search.

Webhook — an HTTP callback Twilio invokes on call events to hand control to our backend.

03 — FEATURE WALKTHROUGH

Every screen: use case, mechanics, value, controls

Each card below covers one page of the app — the CSR scenario it exists for, how it works technically, why it matters to the business, and every button on the screen.

Dashboard — AI Workforce Overview

/dashboard

USE CASE A contact-center manager opens the app each morning to see, at a glance, how the AI workforce performed over the last 14 days — call volume, how many issues it closed without a human, and what that's worth in dollars.

HOW IT WORKS The summary endpoint aggregates Postgres call records: counts by status, resolution/escalation rates as a percentage of completed calls, a rolling 14-day daily call-volume series, and the 8 most recent conversations joined with agent, intent, sentiment, and resolution.

VALUE One screen answers "is the AI workforce keeping up, and is it saving money" without opening a single call record.

CONTROL	WHAT IT DOES
Stat tiles	Active calls, AI agents running, Resolution rate, Cost saved, Total calls, Appointments booked, Human transfer rate, Avg handle time — read-only KPIs (formulas in Section 5).
Call-volume chart	14-day daily call count; hover a bar for the exact figure.
Recent conversations	Latest 8 calls with from-number, agent, intent, sentiment, and resolution badge.

Agent Studio — build & manage AI employees

/agents

USE CASE A supervisor wants a dedicated AI CSR for the pharmacy-benefits queue — speaking Hindi and English, with its own persona and its own plan documents to ground answers in — separate from the default agent that answers the main line.

HOW IT WORKS Each "AI employee" is an `Agent` record with a name, a persona prompt, a mapped TTS voice, and a department. "general" answers directly and routes callers to specialists by detected intent. The per-agent knowledge base panel accepts drag-and-drop PDFs/DOCX; each upload is chunked, embedded, and indexed so `search_documents` can cite it mid-call.

VALUE Different lines of business get an AI CSR trained on their own documents and tone — swap the persona and knowledge base instead of retraining a model.

CONTROL	WHAT IT DOES
Search	Filters the agent grid by name or persona text.
Preview	Plays the selected TTS voice in-browser (approximate stand-in — see Beta notes) before committing to it.
Create	Provisions a new AI employee with the name/voice/persona/department entered.
Duplicate	Clones an existing agent's configuration as a starting point for a new one.
Delete	Permanently removes an agent, after a confirmation dialog.
Save changes	Persists edits to an agent's name, voice, persona, or department.
Make default	Promotes this agent to the one that answers the main inbound line.
Browse files / drag-drop	Uploads knowledge-base documents; shows a per-file progress bar while chunking/embedding.
Delete (document)	Removes an uploaded document and its indexed chunks from the knowledge base.

Live Operations — the supervisor cockpit

/live

USE CASE

A supervisor is watching the floor the way they would a human team. Thirty calls are live, and one caller's confidence score just dropped to 62% because the AI is unsure about a prior-authorization edge case — the supervisor needs to see what's happening and step in without dropping the call.

HOW IT WORKS

Polls in-progress calls every 4 seconds with a live transcript preview and the latest per-turn confidence score, color-banded (good ≥90%, warning 70–89%, critical <70%). "Stop AI" flips the call's `ai_paused` flag, checked before every AI response — the AI goes silent and a human is expected to take over, without dropping the underlying Twilio call. "Send suggestion" writes a coaching note the AI reads on its next turn, like a whisper-coach, without taking the AI fully offline.

VALUE

This is the safety net that makes autonomous calling acceptable in a regulated, health-adjacent domain — nothing runs fully unsupervised, and a human can intervene in seconds.

CONTROL	WHAT IT DOES
Monitor	Opens the live call-detail drawer — transcript, tool calls, and QA-so-far — for that call.
Stop AI / Resume AI	Toggles human-takeover mode for that specific call.
Send suggestion	Opens an inline box to send a real-time coaching prompt to the AI mid-call.

Conversations — Calls & Appointments

/calls

USE CASE

A QA analyst needs to review why a call was escalated, or a compliance reviewer needs the exact transcript of what the AI told a member about their drug coverage.

HOW IT WORKS

Every call is listed with from/to number, handling agent, and detected intent. Opening one shows status, resolution, sentiment, AI confidence, the LLM-generated call summary, the four-axis QA score (accuracy / compliance / empathy / resolution, each independently judged with written notes), and the full turn-by-turn transcript interleaved with tool-call traces and RAG citations.

VALUE

Every AI decision is traceable to the exact data it used — the audit trail a health-plan compliance team will ask for the first time an answer is questioned.

CONTROL	WHAT IT DOES
Status / Agent / Date filters	Narrows the call list.
Row click	Opens the call-detail drawer for that call.
Download .txt NEW	Exports the open call as a plain-text file: metadata, a dedicated sentiment-analysis section (overall sentiment, AI confidence, empathy score), the summary, QA scores, and the full transcript — for offline review or an audit trail.
Live Calls / Appointments tabs	Switches between the call log and booked appointments.

Customers — what the AI remembers

/customers

USE CASE A returning caller shouldn't have to re-verify their name or repeat their preferred language every time they call.

HOW IT WORKS Customer profiles are matched by phone number and enriched during calls by the `update_customer` tool (name, language) plus a rolling `last_interaction` timestamp; profiles can also be bulk-loaded via Data Import.

VALUE Continuity across calls — the difference between a caller feeling remembered versus interrogated from scratch every time.

CONTROL	WHAT IT DOES
Search	Filters by name or phone number.
Row click	Opens the customer detail drawer — profile plus interaction history.

Workflows — deterministic procedures for the AI

/workflows

USE CASE "Whenever a caller wants to change their preferred pharmacy, always verify the member first, then look up the new pharmacy, then confirm the change out loud" — a scripted procedure the AI should follow exactly, not improvise.

HOW IT WORKS A workflow has a plain-language trigger description matched against caller intent, an optional department scope, and an ordered list of steps — each step names one of the 12 registered tools plus an optional natural-language condition (e.g. "only if member is verified"). An active, matching workflow becomes a checklist the AI follows for that call.

VALUE Bridges "the LLM decides what to do" with "compliance requires this exact sequence every time" — the parts of a call that must be scripted, can be.

CONTROL	WHAT IT DOES
Create workflow	Defines a new workflow by name, trigger phrase, and department.
Activate / Deactivate	Turns a workflow on/off for live calls without deleting it.
Delete	Permanently removes a workflow, after confirmation.
+ Add step	Appends a tool call to the workflow's sequence.
↑ / ↓	Reorders a step earlier or later in the sequence.
Remove	Deletes a single step.
Save changes	Persists the workflow's name, trigger, department, active state, and steps.

Analytics — trends across the workspace

/analytics

USE CASE A team lead wants to know if escalations are trending up this week, or which caller intents are dominating call volume before staffing the next shift.

HOW IT WORKS Aggregates the same 14-day window into top intents, sentiment mix, a daily resolved/escalated/abandoned trend, and a daily positive/neutral/negative/frustrated sentiment trend.

VALUE Turns individual call records into a pattern a manager can act on — e.g. staff a pharmacy specialist if "prior authorization" intent is spiking.

CONTROL	WHAT IT DOES
Chart tooltips	Hover any point/bar for the exact daily count behind it.

AI Training Center — the self-improvement loop

/training

USE CASE The AI keeps giving shaky answers about a new drug-tier change that isn't in its knowledge base yet — someone needs to catch that pattern before it shows up in ten more calls.

HOW IT WORKS "Run analysis" triggers an LLM pass over recent calls with low confidence scores, escalations, and low QA scores, clustering them into insights tagged missing_knowledge, prompt_improvement, or workflow_improvement, each backed by a count of supporting example calls.

VALUE Closes the loop from "the AI got it wrong" to "here's exactly what to upload or fix," instead of a supervisor manually reading hundreds of transcripts.

CONTROL	WHAT IT DOES
Run analysis	Kicks off a fresh scan of recent calls for improvement patterns.
All / New / Acknowledged / Dismissed	Filters insights by review status.
Acknowledge	Marks an insight as accepted / being actioned.
Dismiss	Marks an insight as not relevant.

Data Import — bulk-loading real operational data

/data-import

USE CASE Before go-live, the team needs to load the actual member roster, claims history, drug formulary, and pharmacy network from the health plan's systems — not type it in by hand.

HOW IT WORKS Upload a CSV, pick what it represents (members / claims / formulary / pharmacy), map CSV columns to the expected fields (required ones marked), preview the first 5 mapped rows, then import. Re-importing the same file upserts existing records by their natural key instead of duplicating them.

VALUE Gets the AI workforce grounded in real plan data — not fixtures — in minutes, and keeps it in sync as the health plan's own extracts change.

CONTROL	WHAT IT DOES
What are you importing?	Selects the target entity: members, claims, formulary, or pharmacy.
CSV file picker	Loads and parses the CSV, reading its header row.
Column mapping	One dropdown per field, mapping a CSV header to the field it represents.
Import N row(s)	Runs the import and reports created / updated / error counts per row.

04 — THE ACTION REGISTRY

What the AI can actually do on a call

Twelve tools, callable mid-conversation by the LLM's own judgment (or scripted into a Workflow). Every call that uses one logs the full tool-call trace in the transcript.

Tool	What it does	CSR moment it's used in
`search_documents`	RAG lookup over the agent's uploaded knowledge base; returns cited passages.	"What's our policy on 90-day mail-order refills?"
`verify_member`	Confirms caller identity against member records.	Start of every sensitive call, before disclosing PHI.
`check_claim_status`	Looks up the adjudication status of a submitted claim.	"Why hasn't my claim been paid yet?"
`get_benefits`	Retrieves plan benefit details.	"What's my deductible this year?"
`search_formulary`	Looks up a drug's tier, copay, and prior-authorization requirement.	"Is Ozempic covered, and does it need prior auth?"
`find_pharmacy`	Finds in-network pharmacies near a ZIP code.	"Where can I fill this near 110001?"
`schedule_appointment`	Books an appointment/consultation.	"Can someone call me back tomorrow at 3pm?"
`schedule_callback`	Queues a simple callback without a full appointment.	Caller is mid-errand and wants a call later.
`escalate_to_human`	Flags the call for live-agent handoff and sets resolution_status = escalated.	AI can't resolve it, or the caller demands a human.
`create_ticket`	Opens a support ticket for follow-up work.	Issue needs backend research after the call.
`send_email`	Sends a confirmation/summary email to the member.	"Can you email me that?"
`update_customer`	Writes back learned profile facts (name, language).	A first-time caller gives their name.

05 — COST & RESOLUTION ECONOMICS

The numbers behind the dashboard tiles

Exactly how "Resolution rate," "Human transfer rate," "Avg handle time," and "Cost saved" are computed — not marketing round numbers.

Resolution rate

```
resolution_rate = resolved_calls ÷ completed_calls × 100
```

The share of finished calls the AI closed on its own, with no human involved. This is the platform's headline first-call-resolution (FCR) equivalent.

Human transfer rate (escalation rate)

```
escalation_rate = escalated_calls ÷ completed_calls × 100
```

The counterpart metric supervisors watch alongside resolution rate — how often the AI correctly recognized it should hand off, via `escalate_to_human`, rather than force an answer.

Average handle time (AHT)

```
avg_handle_time = mean(ended_at − started_at) over completed calls
```

The same operational metric a contact center already tracks for human agents, so AI and human performance are directly comparable.

Cost saved (estimate)

```
cost_saved_estimate = resolved_calls × $12.00
```

$12 is a stated benchmark assumption for a fully-loaded human-handled contact-center call — a backend constant, not a measurement, and the dashboard labels it as an estimate rather than exact accounting.

Beta note: treat the $12/call figure as a placeholder. Swap it for your own blended cost-per-contact once finance provides one, and the dollar figure on every dashboard recalculates automatically — no other change required.

QA score vs. resolution status — two different judgments

`resolution_status` (resolved / escalated / abandoned) is set procedurally by what actually happened on the call — did a human get pulled in. The four QA scores (accuracy, compliance, empathy, resolution — each 0–100) are an independent LLM-as-judge review of the full transcript and tool-call trace. A call can be "resolved" but still get docked on empathy for a curt tone, or "escalated" appropriately and still score well on compliance for correctly refusing to disclose data before member verification.

06 — DATA HANDLING

Security & access, as currently built

- Multi-tenant by design — every call, agent, customer, and document is scoped to a `tenant_id`.
- Frontend and API access is JWT-authenticated; there is no anonymous access to conversation data.
- Every tool call and message the AI makes is written to the transcript, so any AI decision — and the data it used to make it — is reviewable after the fact.
- Supervisor overrides (Stop AI, Send suggestion) are logged against the call they were issued on.

Beta note: this build has not undergone a formal security or regulatory compliance certification (e.g. HIPAA). Treat it as a pilot environment for evaluation, not yet a certified system for handling production PHI, until that review is complete.

This is a Beta release

Everything described in this document reflects the current build's real, working behavior — nothing here is aspirational. But the platform is still young, and it will keep changing based on what pilot usage surfaces.

07 — KNOWN LIMITATIONS

- **Voice preview is approximate.** The "Preview" button in Agent Studio plays the selected voice via the browser's built-in text-to-speech, as a stand-in for how the production call voice will actually sound.
- **Cost-saved is a placeholder estimate.** It uses a single flat $12/call assumption workspace-wide rather than a per-tenant, per-call-type benchmark.
- **Recent-conversations list isn't click-through yet** on the Dashboard — use the Conversations page for the full call-detail drawer.
- **No formal compliance certification yet** — see Section 6.

WHAT'S NEXT

- Replace the flat cost-saved assumption with a configurable, per-tenant cost-per-contact benchmark.
- Deeper analytics drill-down (click a trend point through to the underlying calls).
- Richer workflow step conditions and branching.
- Formal security and compliance review ahead of any production PHI handling.

Feedback from this pilot directly shapes what gets built next. Thanks for trying the beta — expect this document, and the product, to keep improving.